

Exploratory Data Analysis (EDA) on Titanic Dataset

Objective:

To perform Exploratory Data Analysis (EDA) on the Titanic dataset using Python libraries such as Pandas, Matplotlib, and Seaborn in order to identify patterns, relationships, and insights from the data.

Dataset Overview:

- 1 Dataset contains passenger information from the Titanic disaster.
- 2 Includes numerical and categorical data.
- 3 Features include Age, Fare, Pclass, Sex, Survival status, etc.

Steps Performed:

- 1 Loaded dataset using pandas
- 2 Inspected dataset structure using info() and describe()
- 3 Checked and handled missing values
- 4 Performed distribution analysis using histograms and countplots
- 5 Created correlation heatmap for numerical features
- 6 Detected outliers using boxplots and Z-score method
- 7 Performed advanced visualization using Pairplot and FacetGrid
- 8 Saved cleaned dataset as titanic_cleaned.csv

Tools & Libraries Used:

- 1 Python
- 2 Pandas
- 3 NumPy
- 4 Matplotlib
- 5 Seaborn
- 6 SciPy

Key Insights:

- 1 Female passengers had a higher survival rate than males.
- 2 Passengers in 1st class survived more compared to 3rd class.
- 3 Fare showed a positive relationship with survival.
- 4 Most passengers were between 20–40 years old.
- 5 Some outliers were present in Fare column.

Conclusion:

The EDA process helped in understanding passenger demographics, survival patterns, and feature relationships within the Titanic dataset. The analysis provided valuable insights useful for future predictive modeling and data-driven decision making.

Possible Viva Questions:

- 1 What is Exploratory Data Analysis (EDA)?
- 2 Why are missing values handled?
- 3 What is correlation?
- 4 What are outliers?
- 5 Difference between histogram and boxplot?